
MODEL SELECTION AND VALIDATION REPORT (fishing_kde_fric model)

OBSERVATIONS:1739

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (fishing_kde_fric model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: student(link = 'identity')

Conditional model:

	coef.est	coef.se
(Intercept)	-0.16	0.12
log_sum_fishing	-0.04	0.02
sbt_mean	0.12	0.06
sbt_sd	0.02	0.03
sst_sd	0.05	0.03
sic_mean	-0.04	0.05
log_chla_mean	-0.01	0.03
depth	-0.11	0.03

Dispersion parameter: 0.57

Spatiotemporal AR1 correlation (rho): 0.90

Matérn range: 242.32

Spatiotemporal marginal AR1 SD: 0.51

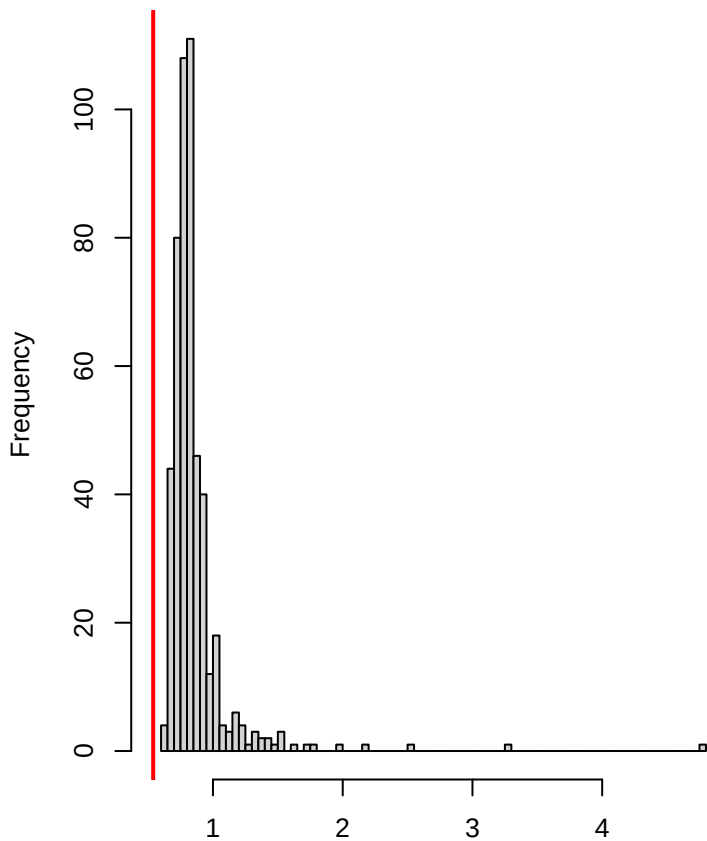
REML criterion at convergence: 2234.592

See ?tidy.sdmTMB to extract these values as a data frame.

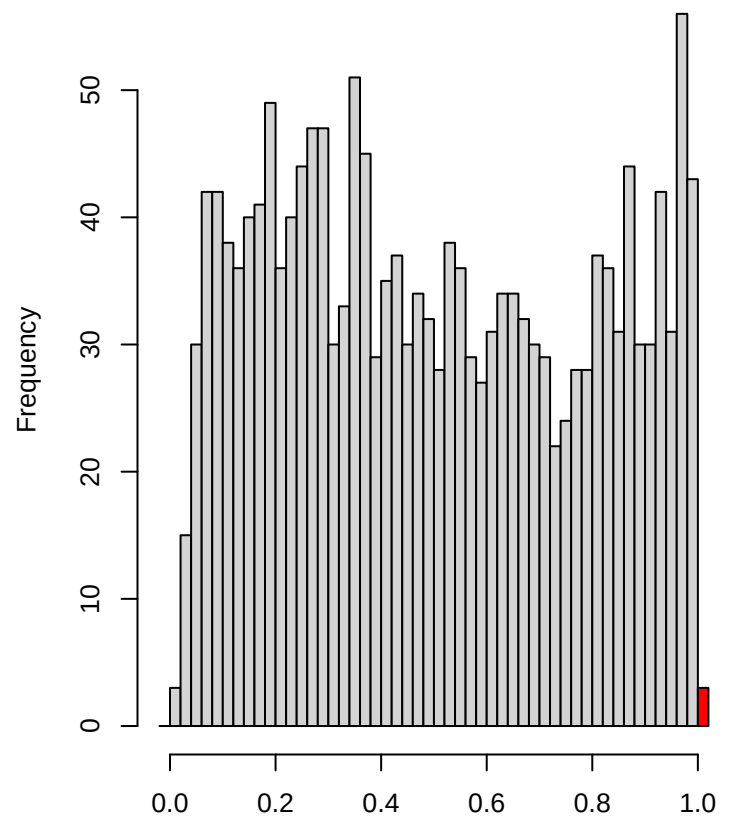
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	12	4493.185	1.162225
spatiotemporal_random_fields_ar1_with_anisotropy	14	4494.347	35.225601
spatial_random_fields	11	4529.572	3.496279
spatial_random_fields_with_anisotropy	13	4533.069	52.867913
spatiotemporal_random_fields_iid_with_anisotropy	13	4585.937	2.047480
spatiotemporal_random_fields_iid	11	4587.984	71.728014
error_only	9	4659.712	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



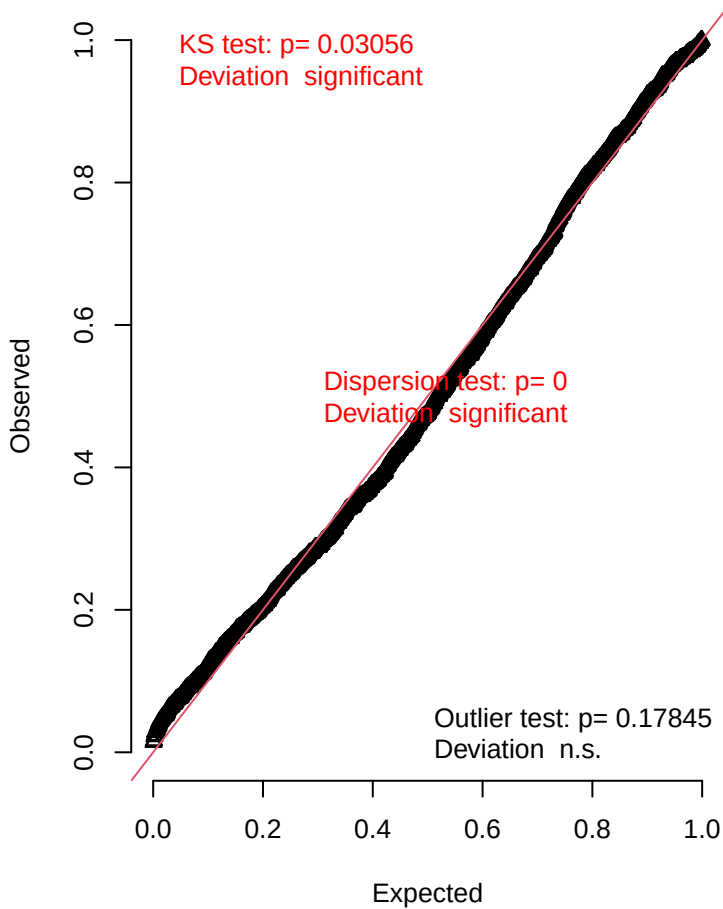
Outlier test n.s.



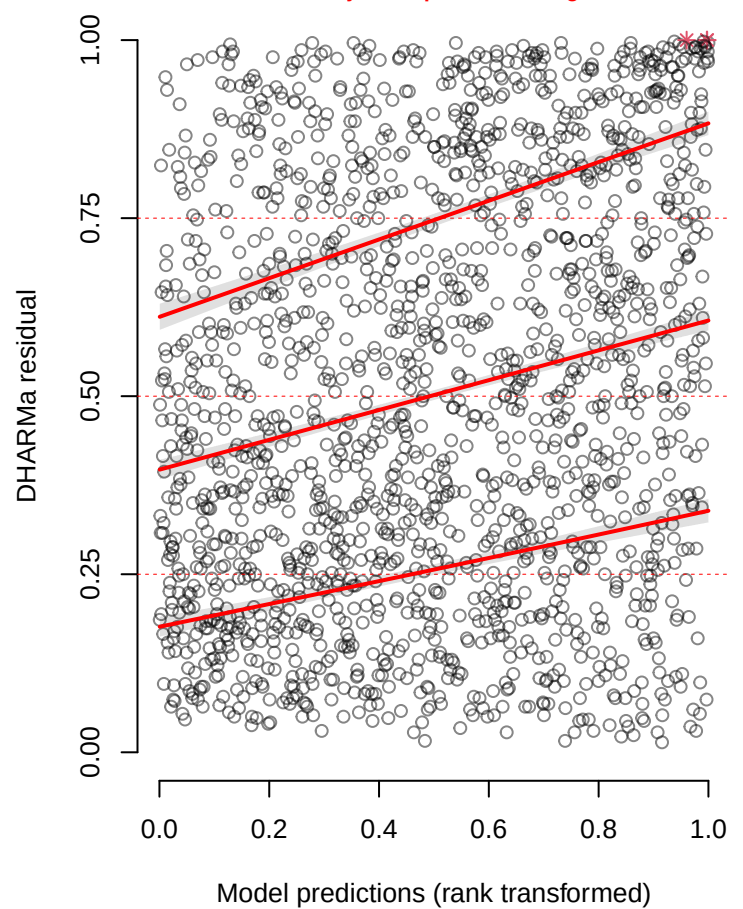
Simulated values, red line = fitted model. p-value (two.sided)

Residuals (outliers are marked red)

QQ plot residuals

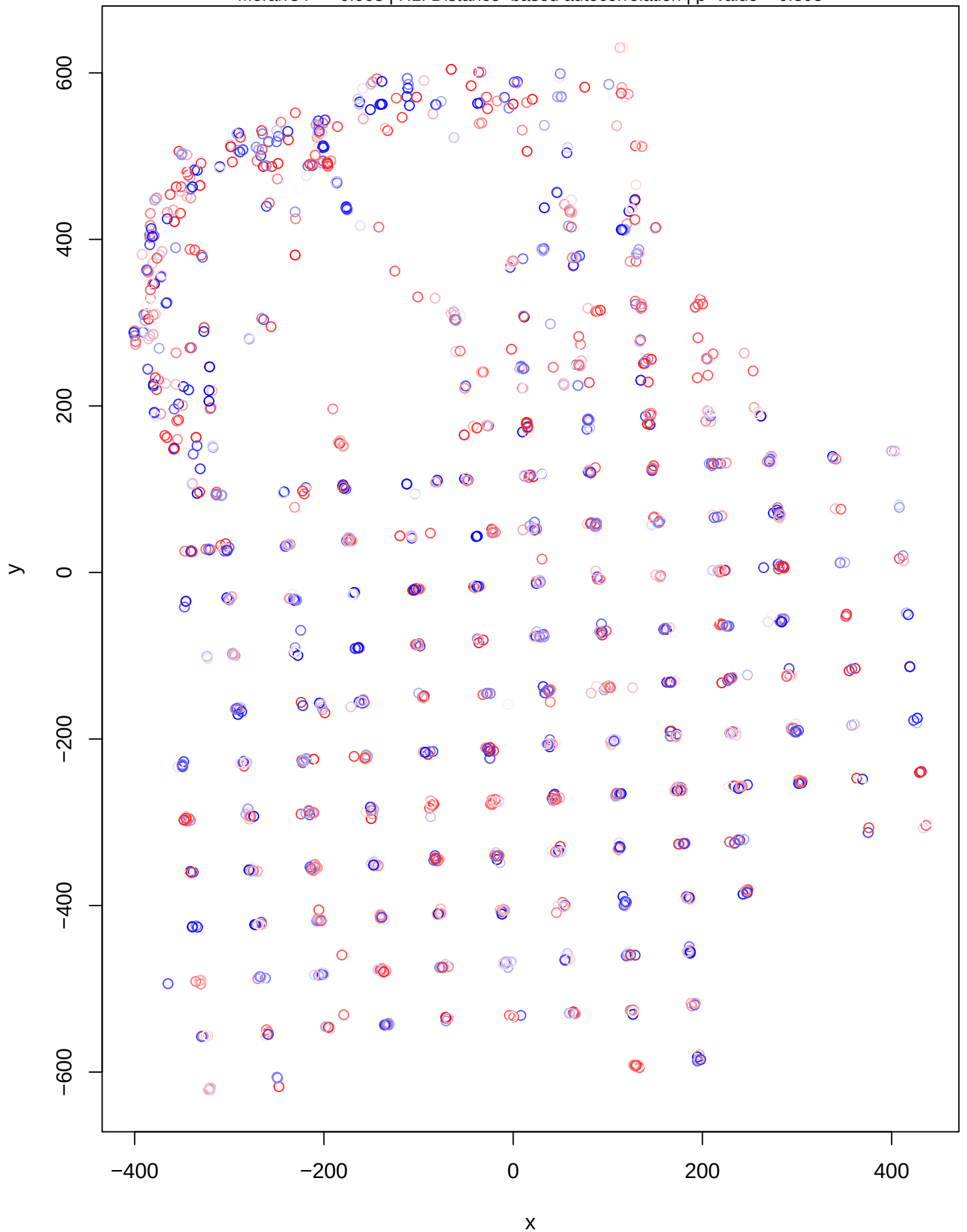


DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant

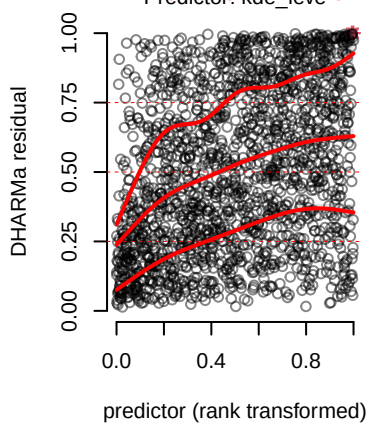


DHARMA Moran's I test for distance-based autocorrelation

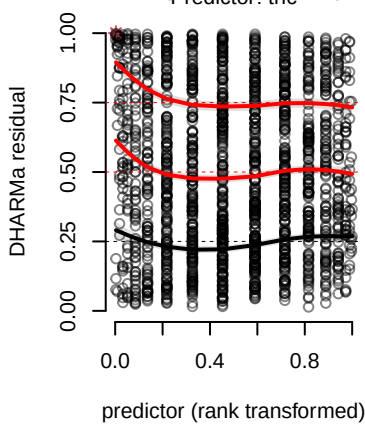
Moran's I = -0.003 | H1: Distance-based autocorrelation | p-value = 0.808



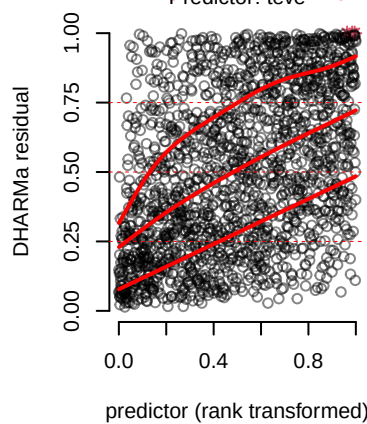
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



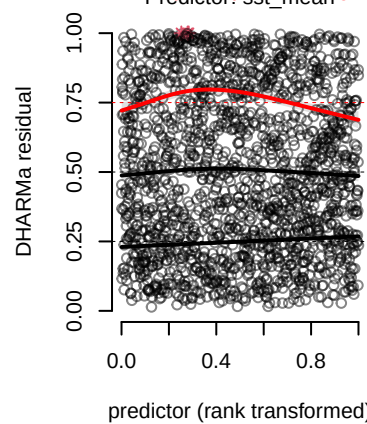
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



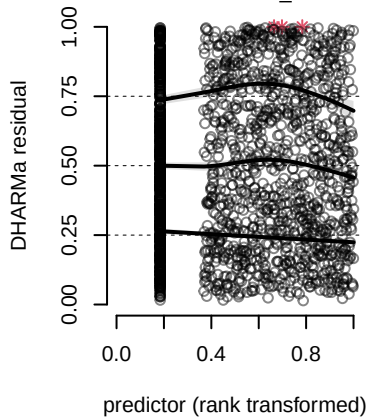
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



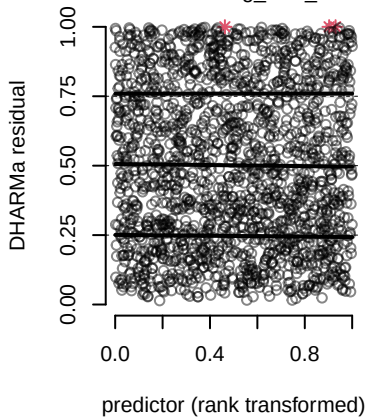
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



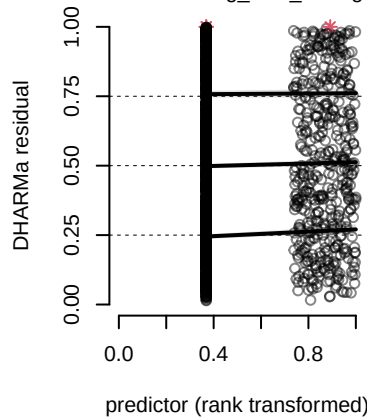
DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: sic_sd



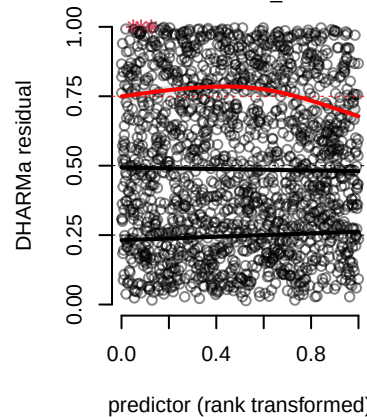
DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: log_chla_sd



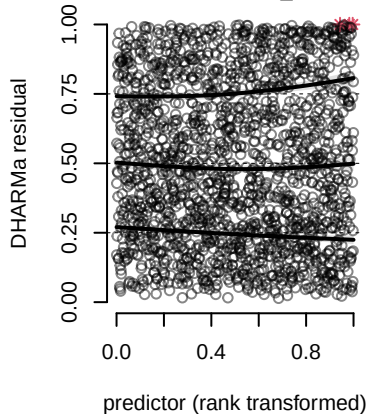
DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: log_sum_fishing



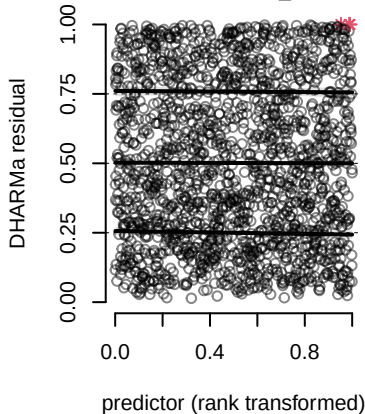
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



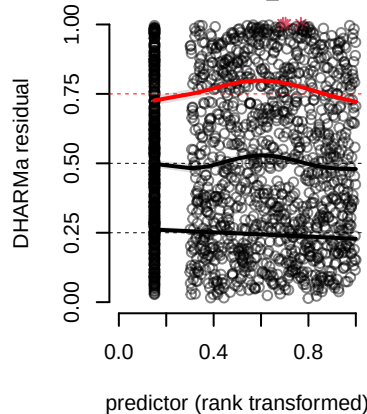
DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: sbt_sd



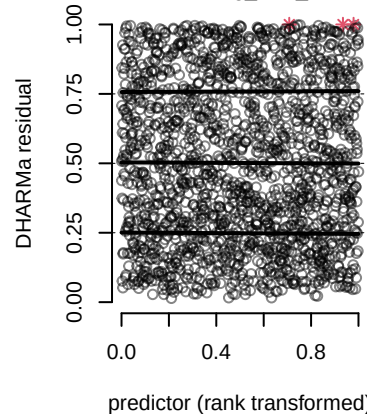
DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: sst_sd



DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: log_chla_mean



DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: depth

